

Topic 1 - Question Set 1

Question #1

Topic 1

Your company has decided to make a major revision of their API in order to create better experiences for their developers. They need to keep the old version of the API available and deployable, while allowing new customers and testers to try out the new API. They want to keep the same SSL and DNS records in place to serve both APIs. What should they do?

- A. Configure a new load balancer for the new version of the API
- B. Reconfigure old clients to use a new endpoint for the new API
- C. Have the old API forward traffic to the new API based on the path
- D. Use separate backend pools for each API path behind the load balancer **Most Voted**

Hide Solution

Discussion 81

Correct Answer: D 

Community vote distribution

D (100%)

Question #2

Topic 1

Your company plans to migrate a multi-petabyte data set to the cloud. The data set must be available 24hrs a day. Your business analysts have experience only with using a SQL interface.

How should you store the data to optimize it for ease of analysis?

- A. Load data into Google BigQuery **Most Voted**
- B. Insert data into Google Cloud SQL
- C. Put flat files into Google Cloud Storage
- D. Stream data into Google Cloud Datastore

Hide Solution

Discussion 80

Correct Answer: A 

BigQuery is Google's serverless, highly scalable, low cost enterprise data warehouse designed to make all your data analysts productive. Because there is no infrastructure to manage, you can focus on analyzing data to find meaningful insights using familiar SQL and you don't need a database administrator.

BigQuery enables you to analyze all your data by creating a logical data warehouse over managed, columnar storage as well as data from object storage, and spreadsheets.

Reference:

<https://cloud.google.com/bigquery/>

Community vote distribution

A (94%)

6%

Question #3

Topic 1

The operations manager asks you for a list of recommended practices that she should consider when migrating a J2EE application to the cloud. Which three practices should you recommend? (Choose three.)

A. Port the application code to run on Google App Engine **Most Voted** **Most Voted**

B. Integrate Cloud Dataflow into the application to capture real-time metrics

C. Instrument the application with a monitoring tool like Stackdriver Debugger **Most Voted** **Most Voted**

D. Select an automation framework to reliably provision the cloud infrastructure **Most Voted** **Most Voted**

E. Deploy a continuous integration tool with automated testing in a staging environment **Most Voted** **Most Voted** **Most Voted**

F. Migrate from MySQL to a managed NoSQL database like Google Cloud Datastore or Bigtable

[Hide Solution](#)[Discussion](#) 114

Correct Answer: ADE 

References:

<https://cloud.google.com/appengine/docs/standard/java/tools/uploadinganapp> <https://cloud.google.com/appengine/docs/standard/java/building-app/cloud-sql>

Community vote distribution

ADE (38%)

ACE (33%)

CDE (29%)

A news feed web service has the following code running on Google App Engine. During peak load, users report that they can see news articles they already viewed. What is the most likely cause of this problem?

```
import news
from flask import Flask, redirect, request
from flask.ext.api import status
from google.appengine.api import users

app = Flask(__name__)
sessions = {}

@app.route("/")
def homepage():
    user = users.get_current_user()
    if not user:
        return "Invalid login",
        status.HTTP_401_UNAUTHORIZED

    if user not in sessions:
        sessions[user] = {"viewed": []}

    news_articles = news.get_new_news (user, sessions [user]
["viewed"])
    sessions [user] ["viewed"] += [n["id"] for n
in news_articles]

    return news.render(news_articles)

if __name__ == "__main__":
    app.run()
```

- A. The session variable is local to just a single instance **Most Voted**
- B. The session variable is being overwritten in Cloud Datastore
- C. The URL of the API needs to be modified to prevent caching
- D. The HTTP Expires header needs to be set to -1 stop caching

[Hide Solution](#)[Discussion](#) 74

Correct Answer: A 

Community vote distribution

A (100%)

Question #5

Topic 1

An application development team believes their current logging tool will not meet their needs for their new cloud-based product. They want a better tool to capture errors and help them analyze their historical log data. You want to help them find a solution that meets their needs.

What should you do?

- A. Direct them to download and install the Google StackDriver logging agent **Most Voted**
- B. Send them a list of online resources about logging best practices
- C. Help them define their requirements and assess viable logging tools **Most Voted**
- D. Help them upgrade their current tool to take advantage of any new features

Hide Solution

Discussion 150

Correct Answer: C 

Community vote distribution

A (52%)

C (48%)

Question #6

Topic 1

You need to reduce the number of unplanned rollbacks of erroneous production deployments in your company's web hosting platform. Improvement to the QA/Test processes accomplished an 80% reduction.

Which additional two approaches can you take to further reduce the rollbacks? (Choose two.)

A. Introduce a green-blue deployment model **Most Voted**

B. Replace the QA environment with canary releases

C. Fragment the monolithic platform into microservices **Most Voted**

D. Reduce the platform's dependency on relational database systems

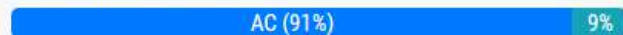
E. Replace the platform's relational database systems with a NoSQL database

Hide Solution

Discussion 37

Correct Answer: AC 

Community vote distribution



Question #7

Topic 1

To reduce costs, the Director of Engineering has required all developers to move their development infrastructure resources from on-premises virtual machines (VMs) to Google Cloud Platform. These resources go through multiple start/stop events during the day and require state to persist. You have been asked to design the process of running a development environment in Google Cloud while providing cost visibility to the finance department.

Which two steps should you take? (Choose two.)

A. Use the `--no-auto-delete` flag on all persistent disks and stop the VM **Most Voted**

B. Use the `--auto-delete` flag on all persistent disks and terminate the VM

C. Apply VM CPU utilization label and include it in the BigQuery billing export

D. Use Google BigQuery billing export and labels to associate cost to groups **Most Voted**

E. Store all state into local SSD, snapshot the persistent disks, and terminate the VM

F. Store all state in Google Cloud Storage, snapshot the persistent disks, and terminate the VM

[Hide Solution](#)[Discussion](#) **60**

Correct Answer: AD 

Community vote distribution

AD (64%)

DF (36%)

Your company wants to track whether someone is present in a meeting room reserved for a scheduled meeting. There are 1000 meeting rooms across 5 offices on 3 continents. Each room is equipped with a motion sensor that reports its status every second. The data from the motion detector includes only a sensor ID and several different discrete items of information. Analysts will use this data, together with information about account owners and office locations.

Which database type should you use?

A. Flat file

B. NoSQL **Most Voted**

C. Relational

D. Blobstore

Hide Solution

Discussion 46

Correct Answer: B 

Relational databases were not designed to cope with the scale and agility challenges that face modern applications, nor were they built to take advantage of the commodity storage and processing power available today.

NoSQL fits well for:

👁️ Developers are working with applications that create massive volumes of new, rapidly changing data types æ" structured, semi-structured, unstructured and polymorphic data.

Incorrect Answers:

D: The Blobstore API allows your application to serve data objects, called blobs, that are much larger than the size allowed for objects in the Datastore service.

Blobs are useful for serving large files, such as video or image files, and for allowing users to upload large data files.

Reference:

<https://www.mongodb.com/nosql-explained>

Community vote distribution

B (100%)

You set up an autoscaling instance group to serve web traffic for an upcoming launch. After configuring the instance group as a backend service to an HTTP(S) load balancer, you notice that virtual machine (VM) instances are being terminated and re-launched every minute. The instances do not have a public IP address. You have verified the appropriate web response is coming from each instance using the curl command. You want to ensure the backend is configured correctly. What should you do?

- A. Ensure that a firewall rule exists to allow source traffic on HTTP/HTTPS to reach the load balancer.
- B. Assign a public IP to each instance and configure a firewall rule to allow the load balancer to reach the instance public IP.
- C. Ensure that a firewall rule exists to allow load balancer health checks to reach the instances in the instance group. **Most Voted**
- D. Create a tag on each instance with the name of the load balancer. Configure a firewall rule with the name of the load balancer as the source and the instance tag as the destination.

[Hide Solution](#)[Discussion](#) 40**Correct Answer:** C 

The best practice when configuring a health check is to check health and serve traffic on the same port. However, it is possible to perform health checks on one port, but serve traffic on another. If you do use two different ports, ensure that firewall rules and services running on instances are configured appropriately. If you run health checks and serve traffic on the same port, but decide to switch ports at some point, be sure to update both the backend service and the health check.

Backend services that do not have a valid global forwarding rule referencing it will not be health checked and will have no health status.

Reference:

<https://cloud.google.com/compute/docs/load-balancing/http/backend-service>

Community vote distribution

C (100%)

Question #10

Topic 1

You write a Python script to connect to Google BigQuery from a Google Compute Engine virtual machine. The script is printing errors that it cannot connect to BigQuery.

What should you do to fix the script?

- A. Install the latest BigQuery API client library for Python
- B. Run your script on a new virtual machine with the BigQuery access scope enabled
- C. Create a new service account with BigQuery access and execute your script with that user **Most Voted**
- D. Install the bq component for gcloud with the command `gcloud components install bq`.

Hide Solution

Discussion 177

Correct Answer: B 

Community vote distribution



Your customer is moving an existing corporate application to Google Cloud Platform from an on-premises data center. The business owners require minimal user disruption. There are strict security team requirements for storing passwords.

What authentication strategy should they use?

- A. Use G Suite Password Sync to replicate passwords into Google
- B. Federate authentication via SAML 2.0 to the existing Identity Provider **Most Voted**
- C. Provision users in Google using the Google Cloud Directory Sync tool
- D. Ask users to set their Google password to match their corporate password

[Hide Solution](#)[Discussion](#) **122**

Correct Answer: C 

Provision users to Google's directory

The global Directory is available to both Cloud Platform and G Suite resources and can be provisioned by a number of means. Provisioned users can take advantage of rich authentication features including single sign-on (SSO), OAuth, and two-factor verification.

You can provision users automatically using one of the following tools and services:

Google Cloud Directory Sync (GCDS)

Google Admin SDK -

A third-party connector -

GCDS is a connector that can provision users and groups on your behalf for both Cloud Platform and G Suite. Using GCDS, you can automate the addition, modification, and deletion of users, groups, and non-employee contacts. You can synchronize the data from your LDAP directory server to your Cloud Platform domain by using LDAP queries. This synchronization is one-way: the data in your LDAP directory server is never modified.

Reference:

<https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations#authentication-and-identity>

Community vote distribution

B (76%)

C (24%)

Your company has successfully migrated to the cloud and wants to analyze their data stream to optimize operations. They do not have any existing code for this analysis, so they are exploring all their options. These options include a mix of batch and stream processing, as they are running some hourly jobs and live- processing some data as it comes in.

Which technology should they use for this?

- A. Google Cloud Dataproc
- B. Google Cloud Dataflow **Most Voted**
- C. Google Container Engine with Bigtable
- D. Google Compute Engine with Google BigQuery

[Hide Solution](#)[Discussion](#) 55

Correct Answer: B 🏆

Cloud Dataflow is a fully-managed service for transforming and enriching data in stream (real time) and batch (historical) modes with equal reliability and expressiveness – no more complex workarounds or compromises needed.

Reference:

<https://cloud.google.com/dataflow/>

Community vote distribution

B (100%)

Your customer is receiving reports that their recently updated Google App Engine application is taking approximately 30 seconds to load for some of their users. This behavior was not reported before the update. What strategy should you take?

- A. Work with your ISP to diagnose the problem
- B. Open a support ticket to ask for network capture and flow data to diagnose the problem, then roll back your application
- C. Roll back to an earlier known good release initially, then use Stackdriver Trace and Logging to diagnose the problem in a development/test/staging environment
Most Voted
- D. Roll back to an earlier known good release, then push the release again at a quieter period to investigate. Then use Stackdriver Trace and Logging to diagnose the problem

[Hide Solution](#)[Discussion](#) 52

Correct Answer: C 

Stackdriver Logging allows you to store, search, analyze, monitor, and alert on log data and events from Google Cloud Platform and Amazon Web Services (AWS). Our API also allows ingestion of any custom log data from any source. Stackdriver Logging is a fully managed service that performs at scale and can ingest application and system log data from thousands of VMs. Even better, you can analyze all that log data in real time.

Reference:

<https://cloud.google.com/logging/>

Community vote distribution

C (91%)

9%

A production database virtual machine on Google Compute Engine has an ext4-formatted persistent disk for data files. The database is about to run out of storage space. How can you remediate the problem with the least amount of downtime?

- A. In the Cloud Platform Console, increase the size of the persistent disk and use the `resize2fs` command in Linux. **Most Voted**
- B. Shut down the virtual machine, use the Cloud Platform Console to increase the persistent disk size, then restart the virtual machine
- C. In the Cloud Platform Console, increase the size of the persistent disk and verify the new space is ready to use with the `fdisk` command in Linux
- D. In the Cloud Platform Console, create a new persistent disk attached to the virtual machine, format and mount it, and configure the database service to move the files to the new disk
- E. In the Cloud Platform Console, create a snapshot of the persistent disk restore the snapshot to a new larger disk, unmount the old disk, mount the new disk and restart the database service

[Hide Solution](#)[Discussion](#) 47

Correct Answer: A 

On Linux instances, connect to your instance and manually resize your partitions and file systems to use the additional disk space that you added.

Extend the file system on the disk or the partition to use the added space. If you grew a partition on your disk, specify the partition. If your disk does not have a partition table, specify only the disk ID. `sudo resize2fs /dev/[DISK_ID][PARTITION_NUMBER]` where [DISK_ID] is the device name and [PARTITION_NUMBER] is the partition number for the device where you are resizing the file system.

Reference:

<https://cloud.google.com/compute/docs/disks/add-persistent-disk>

Community vote distribution

A (100%)

Question #15

Topic 1

Your application needs to process credit card transactions. You want the smallest scope of Payment Card Industry (PCI) compliance without compromising the ability to analyze transactional data and trends relating to which payment methods are used.

How should you design your architecture?

- A. Create a tokenizer service and store only tokenized data **Most Voted**
- B. Create separate projects that only process credit card data
- C. Create separate subnetworks and isolate the components that process credit card data
- D. Streamline the audit discovery phase by labeling all of the virtual machines (VMs) that process PCI data
- E. Enable Logging export to Google BigQuery and use ACLs and views to scope the data shared with the auditor

Hide Solution

Discussion 75

Correct Answer: A 

Reference:

<https://www.sans.org/reading-room/whitepapers/compliance/ways-reduce-pci-dss-audit-scope-tokenizing-cardholder-data-33194>

Community vote distribution

A (100%)

You have been asked to select the storage system for the click-data of your company's large portfolio of websites. This data is streamed in from a custom website analytics package at a typical rate of 6,000 clicks per minute. With bursts of up to 8,500 clicks per second. It must have been stored for future analysis by your data science and user experience teams.

Which storage infrastructure should you choose?

A. Google Cloud SQL

B. Google Cloud Bigtable **Most Voted**

C. Google Cloud Storage

D. Google Cloud Datastore

Hide Solution

Discussion 48

Correct Answer: B 🗳️

Google Cloud Bigtable is a scalable, fully-managed NoSQL wide-column database that is suitable for both real-time access and analytics workloads.

Good for:

- ⇒ Low-latency read/write access
- ⇒ High-throughput analytics
- ⇒ Native time series support

Common workloads:

- ⇒ IoT, finance, adtech
- ⇒ Personalization, recommendations
- ⇒ Monitoring
- ⇒ Geospatial datasets
- ⇒ Graphs

Incorrect Answers:

C: Google Cloud Storage is a scalable, fully-managed, highly reliable, and cost-efficient object / blob store.

Is good for:

- ⇒ Images, pictures, and videos
- ⇒ Objects and blobs
- ⇒ Unstructured data

D: Google Cloud Datastore is a scalable, fully-managed NoSQL document database for your web and mobile applications.

Is good for:

- ⇒ Semi-structured application data
- ⇒ Hierarchical data
- ⇒ Durable key-value data
- ⇒ Common workloads:
- ⇒ User profiles
- ⇒ Product catalogs
- ⇒ Game state

Reference:

<https://cloud.google.com/storage-options/>

Community vote distribution



Question #17

Topic 1

You are creating a solution to remove backup files older than 90 days from your backup Cloud Storage bucket. You want to optimize ongoing Cloud Storage spend. What should you do?

- A. Write a lifecycle management rule in XML and push it to the bucket with gsutil
- B. Write a lifecycle management rule in JSON and push it to the bucket with gsutil **Most Voted**
- C. Schedule a cron script using gsutil ls -l gs://backups/** to find and remove items older than 90 days
- D. Schedule a cron script using gsutil ls -l gs://backups/** to find and remove items older than 90 days and schedule it with cron

Hide Solution

Discussion 43

Correct Answer: B 

Community vote distribution

B (100%)

Your company is forecasting a sharp increase in the number and size of Apache Spark and Hadoop jobs being run on your local datacenter. You want to utilize the cloud to help you scale this upcoming demand with the least amount of operations work and code change.

Which product should you use?

A. Google Cloud Dataflow

B. Google Cloud Dataproc **Most Voted**

C. Google Compute Engine

D. Google Kubernetes Engine

Hide Solution

Discussion 38

Correct Answer: B 

Google Cloud Dataproc is a fast, easy-to-use, low-cost and fully managed service that lets you run the Apache Spark and Apache Hadoop ecosystem on Google Cloud Platform. Cloud Dataproc provisions big or small clusters rapidly, supports many popular job types, and is integrated with other Google Cloud Platform services, such as Google Cloud Storage and Stackdriver Logging, thus helping you reduce TCO.

Reference:

<https://cloud.google.com/dataproc/docs/resources/faq>

Community vote distribution

B (100%)

Question #19

Topic 1

The database administration team has asked you to help them improve the performance of their new database server running on Google Compute Engine. The database is for importing and normalizing their performance statistics and is built with MySQL running on Debian Linux. They have an n1-standard-8 virtual machine with 80 GB of SSD persistent disk.

What should they change to get better performance from this system?

- A. Increase the virtual machine's memory to 64 GB
- B. Create a new virtual machine running PostgreSQL
- C. Dynamically resize the SSD persistent disk to 500 GB **Most Voted**
- D. Migrate their performance metrics warehouse to BigQuery
- E. Modify all of their batch jobs to use bulk inserts into the database

Hide Solution

Discussion 59

Correct Answer: C 

Community vote distribution

C (83%)

Other

You want to optimize the performance of an accurate, real-time, weather-charting application. The data comes from 50,000 sensors sending 10 readings a second, in the format of a timestamp and sensor reading.

Where should you store the data?

A. Google BigQuery

B. Google Cloud SQL

C. Google Cloud Bigtable **Most Voted**

D. Google Cloud Storage

Hide Solution

Discussion 36

Correct Answer: C 

Google Cloud Bigtable is a scalable, fully-managed NoSQL wide-column database that is suitable for both real-time access and analytics workloads.

Good for:

- ☞ Low-latency read/write access
- ☞ High-throughput analytics
- ☞ Native time series support

Common workloads:

- ☞ IoT, finance, adtech
- ☞ Personalization, recommendations
- ☞ Monitoring
- ☞ Geospatial datasets
- ☞ Graphs

Reference:

<https://cloud.google.com/storage-options/>

Community vote distribution

C (100%)

Your company's user-feedback portal comprises a standard LAMP stack replicated across two zones. It is deployed in the us-central1 region and uses autoscaled managed instance groups on all layers, except the database. Currently, only a small group of select customers have access to the portal. The portal meets a 99,99% availability SLA under these conditions. However next quarter, your company will be making the portal available to all users, including unauthenticated users. You need to develop a resiliency testing strategy to ensure the system maintains the SLA once they introduce additional user load.

What should you do?

A. Capture existing users input, and replay captured user load until autoscale is triggered on all layers. At the same time, terminate all resources in one of the zones

Most Voted

B. Create synthetic random user input, replay synthetic load until autoscale logic is triggered on at least one layer, and introduce "chaos" to the system by terminating random resources on both zones

Most Voted

C. Expose the new system to a larger group of users, and increase group size each day until autoscale logic is triggered on all layers. At the same time, terminate random resources on both zones

D. Capture existing users input, and replay captured user load until resource utilization crosses 80%. Also, derive estimated number of users based on existing user's usage of the app, and deploy enough resources to handle 200% of expected load

Hide Solution

Discussion 92

Correct Answer: B 

Community vote distribution

B (79%)

A (21%)

One of the developers on your team deployed their application in Google Container Engine with the Dockerfile below. They report that their application deployments are taking too long.

```
FROM ubuntu:16.04
COPY . /src
RUN apt-get update && apt-get install -y python python-pip
RUN pip install -r requirements.txt
```

You want to optimize this Dockerfile for faster deployment times without adversely affecting the app's functionality.

Which two actions should you take? (Choose two.)

- A. Remove Python after running pip
- B. Remove dependencies from requirements.txt
- C. Use a slimmed-down base image like Alpine Linux **Most Voted**
- D. Use larger machine types for your Google Container Engine node pools
- E. Copy the source after the package dependencies (Python and pip) are installed **Most Voted**

[Hide Solution](#)[Discussion](#) **29**

Correct Answer: CE 

The speed of deployment can be changed by limiting the size of the uploaded app, limiting the complexity of the build necessary in the Dockerfile, if present, and by ensuring a fast and reliable internet connection.

Note: Alpine Linux is built around musl libc and busybox. This makes it smaller and more resource efficient than traditional GNU/Linux distributions. A container requires no more than 8 MB and a minimal installation to disk requires around 130 MB of storage. Not only do you get a fully-fledged Linux environment but a large selection of packages from the repository.

Reference:

<https://groups.google.com/forum/#!topic/google-appengine/hZMEkmmObDU> <https://www.alpinelinux.org/about/>

Community vote distribution

CE (100%)

Question #23

Topic 1

Your solution is producing performance bugs in production that you did not see in staging and test environments. You want to adjust your test and deployment procedures to avoid this problem in the future.

What should you do?

- A. Deploy fewer changes to production
- B. Deploy smaller changes to production
- C. Increase the load on your test and staging environments **Most Voted**
- D. Deploy changes to a small subset of users before rolling out to production

Hide Solution

Discussion 163

Correct Answer: D 

Community vote distribution

C (66%)

D (32%)

Question #24

Topic 1

A small number of API requests to your microservices-based application take a very long time. You know that each request to the API can traverse many services. You want to know which service takes the longest in those cases. What should you do?

- A. Set timeouts on your application so that you can fail requests faster
- B. Send custom metrics for each of your requests to Stackdriver Monitoring
- C. Use Stackdriver Monitoring to look for insights that show when your API latencies are high
- D. Instrument your application with Stackdriver Trace in order to break down the request latencies at each microservice **Most Voted**

Hide Solution

Discussion 37

Correct Answer: D 🗳️

Reference:

https://cloud.google.com/trace/docs/quickstart#find_a_trace

Community vote distribution

D (90%)

10%

Question #25

Topic 1

During a high traffic portion of the day, one of your relational databases crashes, but the replica is never promoted to a master. You want to avoid this in the future. What should you do?

- A. Use a different database
- B. Choose larger instances for your database **Most Voted**
- C. Create snapshots of your database more regularly
- D. Implement routinely scheduled failovers of your databases **Most Voted**

Hide Solution

Discussion 170

Correct Answer: D 

Community vote distribution

D (50%)

B (50%)

Your organization requires that metrics from all applications be retained for 5 years for future analysis in possible legal proceedings. Which approach should you use?

- A. Grant the security team access to the logs in each Project
- B. Configure Stackdriver Monitoring for all Projects, and export to BigQuery **Most Voted**
- C. Configure Stackdriver Monitoring for all Projects with the default retention policies
- D. Configure Stackdriver Monitoring for all Projects, and export to Google Cloud Storage **Most Voted**

[Hide Solution](#)[Discussion](#) 164

Correct Answer: B 

Stackdriver Logging provides you with the ability to filter, search, and view logs from your cloud and open source application services. Allows you to define metrics based on log contents that are incorporated into dashboards and alerts. Enables you to export logs to BigQuery, Google Cloud Storage, and Pub/Sub.

Reference:

<https://cloud.google.com/stackdriver/>

Community vote distribution

D (72%)

B (28%)

Your company has decided to build a backup replica of their on-premises user authentication PostgreSQL database on Google Cloud Platform. The database is 4 TB, and large updates are frequent. Replication requires private address space communication.

Which networking approach should you use?

- A. Google Cloud Dedicated Interconnect **Most Voted**
- B. Google Cloud VPN connected to the data center network
- C. A NAT and TLS translation gateway installed on-premises
- D. A Google Compute Engine instance with a VPN server installed connected to the data center network

[Hide Solution](#)[Discussion](#) 43

Correct Answer: A 

Google Cloud Dedicated Interconnect provides direct physical connections and RFC 1918 communication between your on-premises network and Google's network. Dedicated Interconnect enables you to transfer large amounts of data between networks, which can be more cost effective than purchasing additional bandwidth over the public Internet or using VPN tunnels.

Benefits:

- ☞ Traffic between your on-premises network and your VPC network doesn't traverse the public Internet. Traffic traverses a dedicated connection with fewer hops, meaning there are less points of failure where traffic might get dropped or disrupted.
- ☞ Your VPC network's internal (RFC 1918) IP addresses are directly accessible from your on-premises network. You don't need to use a NAT device or VPN tunnel to reach internal IP addresses. Currently, you can only reach internal IP addresses over a dedicated connection. To reach Google external IP addresses, you must use a separate connection.
- ☞ You can scale your connection to Google based on your needs. Connection capacity is delivered over one or more 10 Gbps Ethernet connections, with a maximum of eight connections (80 Gbps total per interconnect).
- ☞ The cost of egress traffic from your VPC network to your on-premises network is reduced. A dedicated connection is generally the least expensive method if you have a high-volume of traffic to and from Google's network.

Reference:

<https://cloud.google.com/interconnect/docs/details/dedicated>

Community vote distribution

A (100%)

Question #28

Topic 1

Auditors visit your teams every 12 months and ask to review all the Google Cloud Identity and Access Management (Cloud IAM) policy changes in the previous 12 months. You want to streamline and expedite the analysis and audit process.

What should you do?

- A. Create custom Google Stackdriver alerts and send them to the auditor
- B. Enable Logging export to Google BigQuery and use ACLs and views to scope the data shared with the auditor **Most Voted**
- C. Use cloud functions to transfer log entries to Google Cloud SQL and use ACLs and views to limit an auditor's view
- D. Enable Google Cloud Storage (GCS) log export to audit logs into a GCS bucket and delegate access to the bucket

Hide Solution

Discussion 156

Correct Answer: D 

Community vote distribution

B (73%)

D (27%)

Question #29

Topic 1

You are designing a large distributed application with 30 microservices. Each of your distributed microservices needs to connect to a database back-end. You want to store the credentials securely.

Where should you store the credentials?

- A. In the source code
- B. In an environment variable
- C. In a secret management system
- D. In a config file that has restricted access through ACLs

Hide Solution

Discussion 34

Correct Answer: C 

Reference:

<https://cloud.google.com/kms/docs/secret-management>

Community vote distribution

C (100%)

A lead engineer wrote a custom tool that deploys virtual machines in the legacy data center. He wants to migrate the custom tool to the new cloud environment. You want to advocate for the adoption of Google Cloud Deployment Manager. What are two business risks of migrating to Cloud Deployment Manager? (Choose two.)

- A. Cloud Deployment Manager uses Python
- B. Cloud Deployment Manager APIs could be deprecated in the future
- C. Cloud Deployment Manager is unfamiliar to the company's engineers **Most Voted**
- D. Cloud Deployment Manager requires a Google APIs service account to run
- E. Cloud Deployment Manager can be used to permanently delete cloud resources **Most Voted**
- F. Cloud Deployment Manager only supports automation of Google Cloud resources **Most Voted** **Most Voted**

[Hide Solution](#)[Discussion](#) 76

Correct Answer: BF 

Community vote distribution

CF (43%)

CE (26%)

EF (19%)

10%

A development manager is building a new application. He asks you to review his requirements and identify what cloud technologies he can use to meet them. The application must:

1. Be based on open-source technology for cloud portability
2. Dynamically scale compute capacity based on demand
3. Support continuous software delivery
4. Run multiple segregated copies of the same application stack
5. Deploy application bundles using dynamic templates
6. Route network traffic to specific services based on URL

Which combination of technologies will meet all of his requirements?

- A. Google Kubernetes Engine, Jenkins, and Helm **Most Voted**
- B. Google Kubernetes Engine and Cloud Load Balancing
- C. Google Kubernetes Engine and Cloud Deployment Manager
- D. Google Kubernetes Engine, Jenkins, and Cloud Load Balancing

[Hide Solution](#)[Discussion](#) 46

Correct Answer: D 🗳️

Jenkins is an open-source automation server that lets you flexibly orchestrate your build, test, and deployment pipelines. Kubernetes Engine is a hosted version of Kubernetes, a powerful cluster manager and orchestration system for containers.

When you need to set up a continuous delivery (CD) pipeline, deploying Jenkins on Kubernetes Engine provides important benefits over a standard VM-based deployment

Incorrect Answers:

A: Helm is a tool for managing Kubernetes charts. Charts are packages of pre-configured Kubernetes resources.

Use Helm to:

Find and use popular software packaged as Kubernetes charts

•

- 🔗 Share your own applications as Kubernetes charts
- 🔗 Create reproducible builds of your Kubernetes applications
- 🔗 Intelligently manage your Kubernetes manifest files
- 🔗 Manage releases of Helm packages

Reference:

<https://cloud.google.com/solutions/jenkins-on-kubernetes-engine>

Community vote distribution

A (73%)

D (27%)

You have created several pre-emptible Linux virtual machine instances using Google Compute Engine. You want to properly shut down your application before the virtual machines are preempted.

What should you do?

- A. Create a shutdown script named `k99.shutdown` in the `/etc/rc.6.d/` directory
- B. Create a shutdown script registered as a `xinetd` service in Linux and configure a Stackdriver endpoint check to call the service
- C. Create a shutdown script and use it as the value for a new metadata entry with the key `shutdown-script` in the Cloud Platform Console when you create the new virtual machine instance **Most Voted**
- D. Create a shutdown script, registered as a `xinetd` service in Linux, and use the `gcloud compute instances add-metadata` command to specify the service URL as the value for a new metadata entry with the key `shutdown-script-url`

[Hide Solution](#)[Discussion](#) **67**

Correct Answer: C 

A startup script, or a shutdown script, is specified through the metadata server, using startup script metadata keys.

Reference:

<https://cloud.google.com/compute/docs/startupscript>

Community vote distribution

C (68%)

D (32%)

Your organization has a 3-tier web application deployed in the same network on Google Cloud Platform. Each tier (web, API, and database) scales independently of the others. Network traffic should flow through the web to the API tier and then on to the database tier. Traffic should not flow between the web and the database tier. How should you configure the network?

- A. Add each tier to a different subnetwork
- B. Set up software based firewalls on individual VMs
- C. Add tags to each tier and set up routes to allow the desired traffic flow
- D. Add tags to each tier and set up firewall rules to allow the desired traffic flow

[Hide Solution](#)[Discussion](#) 44

Correct Answer: D 

Google Cloud Platform(GCP) enforces firewall rules through rules and tags. GCP rules and tags can be defined once and used across all regions.

Reference:

<https://cloud.google.com/docs/compare/openstack/>

<https://aws.amazon.com/it/blogs/aws/building-three-tier-architectures-with-security-groups/>

Community vote distribution

D (100%)

Your development team has installed a new Linux kernel module on the batch servers in Google Compute Engine (GCE) virtual machines (VMs) to speed up the nightly batch process. Two days after the installation, 50% of the batch servers failed the nightly batch run. You want to collect details on the failure to pass back to the development team.

Which three actions should you take? (Choose three.)

A. Use Stackdriver Logging to search for the module log entries **Most Voted**

B. Read the debug GCE Activity log using the API or Cloud Console

C. Use gcloud or Cloud Console to connect to the serial console and observe the logs **Most Voted**

D. Identify whether a live migration event of the failed server occurred, using in the activity log

E. Adjust the Google Stackdriver timeline to match the failure time, and observe the batch server metrics **Most Voted**

F. Export a debug VM into an image, and run the image on a local server where kernel log messages will be displayed on the native screen

Hide Solution

Discussion 24

Correct Answer: ACE 

Community vote distribution

ACE (100%)

Question #35

Topic 1

Your company wants to try out the cloud with low risk. They want to archive approximately 100 TB of their log data to the cloud and test the analytics features available to them there, while also retaining that data as a long-term disaster recovery backup.

Which two steps should you take? (Choose two.)

A. Load logs into Google BigQuery **Most Voted**

B. Load logs into Google Cloud SQL

C. Import logs into Google Stackdriver

D. Insert logs into Google Cloud Bigtable

E. Upload log files into Google Cloud Storage **Most Voted**

[Hide Solution](#)[Discussion](#) 23

Correct Answer: AE 

Community vote distribution

AE (88%)

13%

You created a pipeline that can deploy your source code changes to your infrastructure in instance groups for self-healing. One of the changes negatively affects your key performance indicator. You are not sure how to fix it, and investigation could take up to a week.

What should you do?

- A. Log in to a server, and iterate on the fix locally
- B. Revert the source code change, and rerun the deployment pipeline **Most Voted**
- C. Log into the servers with the bad code change, and swap in the previous code
- D. Change the instance group template to the previous one, and delete all instances **Most Voted**

[Hide Solution](#)[Discussion](#) **113**

Correct Answer: B 

Community vote distribution

B (59%)

D (40%)

Your organization wants to control IAM policies for different departments independently, but centrally. Which approach should you take?

- A. Multiple Organizations with multiple Folders
- B. Multiple Organizations, one for each department
- C. A single Organization with Folders for each department
- D. A single Organization with multiple projects, each with a central owner

[Hide Solution](#)[Discussion](#) 38

Correct Answer: C 

Folders are nodes in the Cloud Platform Resource Hierarchy. A folder can contain projects, other folders, or a combination of both. You can use folders to group projects under an organization in a hierarchy. For example, your organization might contain multiple departments, each with its own set of GCP resources. Folders allow you to group these resources on a per-department basis. Folders are used to group resources that share common IAM policies. While a folder can contain multiple folders or resources, a given folder or resource can have exactly one parent.

Reference:

<https://cloud.google.com/resource-manager/docs/creating-managing-folders>

Community vote distribution

C (100%)

You deploy your custom Java application to Google App Engine. It fails to deploy and gives you the following stack trace.

What should you do?

```
java.lang.SecurityException: SHA1 digest error for
com/Altostrat/CloakedServlet.class
    at com.google.appengine.runtime.Request.process
-d36f818a24b8cf1d (Request.java)
    at
sun.security.util.ManifestEntryVerifier.verify
(ManifestEntryVerifier.java:210)
    at java.util.jar.JarVerifier.processEntry
(JarVerifier.java:218)
    at java.util.jar.JarVerifier.update
(JarVerifier.java:205)
    at
java.util.jar.JarVerifiersVerifierStream.read
(JarVerifier.java:428)
    at sun.misc.Resource.getBytes
(Resource.java:124)
    at java.net.URL.ClassLoader.defineClass
(URLClassLoader.java:273)
    at sun.reflect.GeneratedMethodAccessor5.invoke
(Unknown Source)
    at
sun.reflect.DelegatingMethodAccessorImpl.invoke
(DelegatingMethodAccessorImpl.java:43)
    at java.lang.reflect.Method.invoke
(Method.java:616)
    at java.lang.ClassLoader.loadClass
(ClassLoader.java:266)
```

- A. Upload missing JAR files and redeploy your application.
- B. Digitally sign all of your JAR files and redeploy your application **Most Voted**
- C. Recompile the CLoakedServlet class using and MD5 hash instead of SHA1

[Hide Solution](#)[Discussion](#) 34

Correct Answer: B 🗳️

Community vote distribution

8 (100%)

Question #40

Topic 1

As part of implementing their disaster recovery plan, your company is trying to replicate their production MySQL database from their private data center to their GCP project using a Google Cloud VPN connection. They are experiencing latency issues and a small amount of packet loss that is disrupting the replication. What should they do?

- A. Configure their replication to use UDP.
- B. Configure a Google Cloud Dedicated Interconnect. **Most Voted**
- C. Restore their database daily using Google Cloud SQL.
- D. Add additional VPN connections and load balance them.
- E. Send the replicated transaction to Google Cloud Pub/Sub.

Hide Solution

Discussion 66

Correct Answer: B 

Community vote distribution

B (100%)

Question #40

Topic 1

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- D. Add additional VPN connections and load balance them.
- E. Send the replicated transaction to Google Cloud Pub/Sub.

Hide Solution

Discussion 66

Correct Answer: B 

Community vote distribution

B (100%)

Question #41

Topic 1

Your customer support tool logs all email and chat conversations to Cloud Bigtable for retention and analysis. What is the recommended approach for sanitizing this data of personally identifiable information or payment card information before initial storage?

- A. Hash all data using SHA256
- B. Encrypt all data using elliptic curve cryptography
- C. De-identify the data with the Cloud Data Loss Prevention API **Most Voted**
- D. Use regular expressions to find and redact phone numbers, email addresses, and credit card numbers

Hide Solution

Discussion 36

Correct Answer: C 🏆

Reference:

https://cloud.google.com/solutions/pci-dss-compliance-in-gcp#using_data_loss_prevention_api_to_sanitize_data

Community vote distribution

C (100%)

Question #42

Topic 1

You are using Cloud Shell and need to install a custom utility for use in a few weeks. Where can you store the file so it is in the default execution path and persists across sessions?

A. ~/bin **Most Voted**

B. Cloud Storage

C. /google/scripts

D. /usr/local/bin

Hide Solution

Discussion 70

Correct Answer: A 

Community vote distribution

A (100%)

Question #43

Topic 1

You want to create a private connection between your instances on Compute Engine and your on-premises data center. You require a connection of at least 20 Gbps. You want to follow Google-recommended practices. How should you set up the connection?

- A. Create a VPC and connect it to your on-premises data center using Dedicated Interconnect. **Most Voted**
- B. Create a VPC and connect it to your on-premises data center using a single Cloud VPN.
- C. Create a Cloud Content Delivery Network (Cloud CDN) and connect it to your on-premises data center using Dedicated Interconnect.
- D. Create a Cloud Content Delivery Network (Cloud CDN) and connect it to your on-premises datacenter using a single Cloud VPN.

Hide Solution

Discussion 38

Correct Answer: A 

Community vote distribution

A (100%)

Question #44

Topic 1

You are analyzing and defining business processes to support your startup's trial usage of GCP, and you don't yet know what consumer demand for your product will be. Your manager requires you to minimize GCP service costs and adhere to Google best practices. What should you do?

- A. Utilize free tier and sustained use discounts. Provision a staff position for service cost management.
- B. Utilize free tier and sustained use discounts. Provide training to the team about service cost management. **Most Voted**
- C. Utilize free tier and committed use discounts. Provision a staff position for service cost management.
- D. Utilize free tier and committed use discounts. Provide training to the team about service cost management.

Hide Solution

Discussion 69

Correct Answer: B 

Community vote distribution

B (100%)

Question #45

Topic 1

You are building a continuous deployment pipeline for a project stored in a Git source repository and want to ensure that code changes can be verified before deploying to production. What should you do?

- A. Use Spinnaker to deploy builds to production using the red/black deployment strategy so that changes can easily be rolled back.
- B. Use Spinnaker to deploy builds to production and run tests on production deployments.
- C. Use Jenkins to build the staging branches and the master branch. Build and deploy changes to production for 10% of users before doing a complete rollout.
- D. Use Jenkins to monitor tags in the repository. Deploy staging tags to a staging environment for testing. After testing, tag the repository for production and deploy that to the production environment. **Most Voted**

Hide Solution

Discussion 98

Correct Answer: D 

Reference:

<https://github.com/GoogleCloudPlatform/continuous-deployment-on-kubernetes/blob/master/README.md>

Community vote distribution

D (100%)

Question #46

Topic 1

You have an outage in your Compute Engine managed instance group: all instances keep restarting after 5 seconds. You have a health check configured, but autoscaling is disabled. Your colleague, who is a Linux expert, offered to look into the issue. You need to make sure that he can access the VMs. What should you do?

- A. Grant your colleague the IAM role of project Viewer
- B. Perform a rolling restart on the instance group
- C. Disable the health check for the instance group. Add his SSH key to the project-wide SSH Keys **Most Voted**
- D. Disable autoscaling for the instance group. Add his SSH key to the project-wide SSH Keys

[Hide Solution](#)[Discussion](#) 113**Correct Answer:** C *Community vote distribution*

C (81%)

A (19%)

Your company is migrating its on-premises data center into the cloud. As part of the migration, you want to integrate Google Kubernetes Engine (GKE) for workload orchestration. Parts of your architecture must also be PCI DSS-compliant. Which of the following is most accurate?

- A. App Engine is the only compute platform on GCP that is certified for PCI DSS hosting.
- B. GKE cannot be used under PCI DSS because it is considered shared hosting.
- C. GKE and GCP provide the tools you need to build a PCI DSS-compliant environment. **Most Voted**
- D. All Google Cloud services are usable because Google Cloud Platform is certified PCI-compliant.

[Hide Solution](#)[Discussion](#) **15**

Correct Answer: C 

Community vote distribution

C (100%)

Question #48

Topic 1

Your company has multiple on-premises systems that serve as sources for reporting. The data has not been maintained well and has become degraded over time. You want to use Google-recommended practices to detect anomalies in your company data. What should you do?

- A. Upload your files into Cloud Storage. Use Cloud Datalab to explore and clean your data.
- B. Upload your files into Cloud Storage. Use Cloud Dataprep to explore and clean your data. **Most Voted**
- C. Connect Cloud Datalab to your on-premises systems. Use Cloud Datalab to explore and clean your data.
- D. Connect Cloud Dataprep to your on-premises systems. Use Cloud Dataprep to explore and clean your data.

[Hide Solution](#)[Discussion](#) 57**Correct Answer:** B *Community vote distribution* B (100%)

Google Cloud Platform resources are managed hierarchically using organization, folders, and projects. When Cloud Identity and Access Management (IAM) policies exist at these different levels, what is the effective policy at a particular node of the hierarchy?

- A. The effective policy is determined only by the policy set at the node
- B. The effective policy is the policy set at the node and restricted by the policies of its ancestors
- C. The effective policy is the union of the policy set at the node and policies inherited from its ancestors
- D. The effective policy is the intersection of the policy set at the node and policies inherited from its ancestors

[Hide Solution](#)[Discussion](#) 37

Correct Answer: C 

Reference:

<https://cloud.google.com/resource-manager/docs/cloud-platform-resource-hierarchy>

Community vote distribution

C (100%)

You are migrating your on-premises solution to Google Cloud in several phases. You will use Cloud VPN to maintain a connection between your on-premises systems and Google Cloud until the migration is completed. You want to make sure all your on-premise systems remain reachable during this period. How should you organize your networking in Google Cloud?

- A. Use the same IP range on Google Cloud as you use on-premises
- B. Use the same IP range on Google Cloud as you use on-premises for your primary IP range and use a secondary range that does not overlap with the range you use on-premises
- C. Use an IP range on Google Cloud that does not overlap with the range you use on-premises **Most Voted**
- D. Use an IP range on Google Cloud that does not overlap with the range you use on-premises for your primary IP range and use a secondary range with the same IP range as you use on-premises

[Hide Solution](#)[Discussion](#) 86

Correct Answer: C 

Community vote distribution

C (100%)

You have found an error in your App Engine application caused by missing Cloud Datastore indexes. You have created a YAML file with the required indexes and want to deploy these new indexes to Cloud Datastore. What should you do?

- A. Point gcloud datastore create-indexes to your configuration file **Most Voted**
- B. Upload the configuration file to App Engine's default Cloud Storage bucket, and have App Engine detect the new indexes
- C. In the GCP Console, use Datastore Admin to delete the current indexes and upload the new configuration file
- D. Create an HTTP request to the built-in python module to send the index configuration file to your application

[Hide Solution](#)[Discussion](#) **71**

Correct Answer: A 

Community vote distribution

A (100%)

You have an application that will run on Compute Engine. You need to design an architecture that takes into account a disaster recovery plan that requires your application to fail over to another region in case of a regional outage. What should you do?

- A. Deploy the application on two Compute Engine instances in the same project but in a different region. Use the first instance to serve traffic, and use the HTTP load balancing service to fail over to the standby instance in case of a disaster.
- B. Deploy the application on a Compute Engine instance. Use the instance to serve traffic, and use the HTTP load balancing service to fail over to an instance on your premises in case of a disaster.
- C. Deploy the application on two Compute Engine instance groups, each in the same project but in a different region. Use the first instance group to serve traffic, and use the HTTP load balancing service to fail over to the standby instance group in case of a disaster. **Most Voted**
- D. Deploy the application on two Compute Engine instance groups, each in a separate project and a different region. Use the first instance group to serve traffic, and use the HTTP load balancing service to fail over to the standby instance group in case of a disaster.

[Hide Solution](#)[Discussion](#) **46**

Correct Answer: C 

Community vote distribution

C (100%)

You are deploying an application on App Engine that needs to integrate with an on-premises database. For security purposes, your on-premises database must not be accessible through the public internet. What should you do?

- A. Deploy your application on App Engine standard environment and use App Engine firewall rules to limit access to the open on-premises database.
- B. Deploy your application on App Engine standard environment and use Cloud VPN to limit access to the on-premises database.
- C. Deploy your application on App Engine flexible environment and use App Engine firewall rules to limit access to the on-premises database. **Most Voted**
- D. Deploy your application on App Engine flexible environment and use Cloud VPN to limit access to the on-premises database. **Most Voted**

[Hide Solution](#)[Discussion](#) **141**

Correct Answer: D 

Community vote distribution



You are working in a highly secured environment where public Internet access from the Compute Engine VMs is not allowed. You do not yet have a VPN connection to access an on-premises file server. You need to install specific software on a Compute Engine instance. How should you install the software?

- A. Upload the required installation files to Cloud Storage. Configure the VM on a subnet with a Private Google Access subnet. Assign only an internal IP address to the VM. Download the installation files to the VM using gsutil. **Most Voted**
- B. Upload the required installation files to Cloud Storage and use firewall rules to block all traffic except the IP address range for Cloud Storage. Download the files to the VM using gsutil.
- C. Upload the required installation files to Cloud Source Repositories. Configure the VM on a subnet with a Private Google Access subnet. Assign only an internal IP address to the VM. Download the installation files to the VM using gcloud.
- D. Upload the required installation files to Cloud Source Repositories and use firewall rules to block all traffic except the IP address range for Cloud Source Repositories. Download the files to the VM using gsutil.

[Hide Solution](#)[Discussion](#) **96**

Correct Answer: A 

Community vote distribution

A (81%)

B (19%)

Question #55

Topic 1

Your company is moving 75 TB of data into Google Cloud. You want to use Cloud Storage and follow Google-recommended practices. What should you do?

- A. Move your data onto a Transfer Appliance. Use a Transfer Appliance Rehydrator to decrypt the data into Cloud Storage. **Most Voted**
- B. Move your data onto a Transfer Appliance. Use Cloud Dataprep to decrypt the data into Cloud Storage.
- C. Install gsutil on each server that contains data. Use resumable transfers to upload the data into Cloud Storage.
- D. Install gsutil on each server containing data. Use streaming transfers to upload the data into Cloud Storage.

Hide Solution

Discussion 130

Correct Answer: A 

Community vote distribution

A (89%)

11%

Question #56

Topic 1

You have an application deployed on Google Kubernetes Engine using a Deployment named echo-deployment. The deployment is exposed using a Service called echo-service. You need to perform an update to the application with minimal downtime to the application. What should you do?

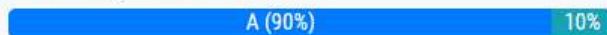
- A. Use `kubectl set image deployment/echo-deployment <new-image>` **Most Voted**
- B. Use the rolling update functionality of the Instance Group behind the Kubernetes cluster
- C. Update the deployment yaml file with the new container image. Use `kubectl delete deployment/echo-deployment` and `kubectl create -f <yaml-file>`
- D. Update the service yaml file with the new container image. Use `kubectl delete service/echo-service` and `kubectl create -f <yaml-file>`

Hide Solution

Discussion 84

Correct Answer: A 

Community vote distribution



Your company is using BigQuery as its enterprise data warehouse. Data is distributed over several Google Cloud projects. All queries on BigQuery need to be billed on a single project. You want to make sure that no query costs are incurred on the projects that contain the data. Users should be able to query the datasets, but not edit them. How should you configure users' access roles?

- A. Add all users to a group. Grant the group the role of BigQuery user on the billing project and BigQuery dataViewer on the projects that contain the data.
- B. Add all users to a group. Grant the group the roles of BigQuery dataViewer on the billing project and BigQuery user on the projects that contain the data.
- C. Add all users to a group. Grant the group the roles of BigQuery jobUser on the billing project and BigQuery dataViewer on the projects that contain the data.
- D. Add all users to a group. Grant the group the roles of BigQuery dataViewer on the billing project and BigQuery jobUser on the projects that contain the data.

Most Voted[Hide Solution](#)[Discussion](#) **16****Correct Answer:** C *Community vote distribution***C (100%)**

Question #58

Topic 1

You have developed an application using Cloud ML Engine that recognizes famous paintings from uploaded images. You want to test the application and allow specific people to upload images for the next 24 hours. Not all users have a Google Account. How should you have users upload images?

- A. Have users upload the images to Cloud Storage. Protect the bucket with a password that expires after 24 hours.
- B. Have users upload the images to Cloud Storage using a signed URL that expires after 24 hours.
- C. Create an App Engine web application where users can upload images. Configure App Engine to disable the application after 24 hours. Authenticate users via Cloud Identity.
- D. Create an App Engine web application where users can upload images for the next 24 hours. Authenticate users via Cloud Identity.

Hide Solution

Discussion 66

Correct Answer: B 

Community vote distribution

B (100%)

Question #59

Topic 1

Your web application must comply with the requirements of the European Union's General Data Protection Regulation (GDPR). You are responsible for the technical architecture of your web application. What should you do?

- A. Ensure that your web application only uses native features and services of Google Cloud Platform, because Google already has various certifications and provides *pass-on* compliance when you use native features.
- B. Enable the relevant GDPR compliance setting within the GCP Console for each of the services in use within your application.
- C. Ensure that Cloud Security Scanner is part of your test planning strategy in order to pick up any compliance gaps.
- D. Define a design for the security of data in your web application that meets GDPR requirements. **Most Voted**

Hide Solution

Discussion 33

Correct Answer: D 

Reference:

<https://www.mobiloud.com/blog/gdpr-compliant-mobile-app/>

Community vote distribution

D (100%)

Question #60

Topic 1

You need to set up Microsoft SQL Server on GCP. Management requires that there's no downtime in case of a data center outage in any of the zones within a GCP region. What should you do?

- A. Configure a Cloud SQL instance with high availability enabled. **Most Voted**
- B. Configure a Cloud Spanner instance with a regional instance configuration.
- C. Set up SQL Server on Compute Engine, using Always On Availability Groups using Windows Failover Clustering. Place nodes in different subnets.
- D. Set up SQL Server Always On Availability Groups using Windows Failover Clustering. Place nodes in different zones.

Hide Solution

Discussion 216

Correct Answer: D 

Community vote distribution



The development team has provided you with a Kubernetes Deployment file. You have no infrastructure yet and need to deploy the application. What should you do?

- A. Use gcloud to create a Kubernetes cluster. Use Deployment Manager to create the deployment.
- B. Use gcloud to create a Kubernetes cluster. Use kubectl to create the deployment. **Most Voted**
- C. Use kubectl to create a Kubernetes cluster. Use Deployment Manager to create the deployment.
- D. Use kubectl to create a Kubernetes cluster. Use kubectl to create the deployment.

[Hide Solution](#)[Discussion](#) **81**

Correct Answer: B 

Community vote distribution

B (100%)

Question #62

Topic 1

You need to evaluate your team readiness for a new GCP project. You must perform the evaluation and create a skills gap plan which incorporates the business goal of cost optimization. Your team has deployed two GCP projects successfully to date. What should you do?

- A. Allocate budget for team training. Set a deadline for the new GCP project.
- B. Allocate budget for team training. Create a roadmap for your team to achieve Google Cloud certification based on job role. **Most Voted**
- C. Allocate budget to hire skilled external consultants. Set a deadline for the new GCP project.
- D. Allocate budget to hire skilled external consultants. Create a roadmap for your team to achieve Google Cloud certification based on job role.

[Hide Solution](#)[Discussion](#) 100**Correct Answer:** A *Community vote distribution*

B (77%)

A (15%)

8%

Question #63

Topic 1

You are designing an application for use only during business hours. For the minimum viable product release, you'd like to use a managed product that automatically `scales to zero` so you don't incur costs when there is no activity.

Which primary compute resource should you choose?

- A. Cloud Functions **Most Voted**
- B. Compute Engine
- C. Google Kubernetes Engine
- D. AppEngine flexible environment

Hide Solution

Discussion 19

Correct Answer: A 

Community vote distribution



Question #64

Topic 1

You are creating an App Engine application that uses Cloud Datastore as its persistence layer. You need to retrieve several root entities for which you have the identifiers. You want to minimize the overhead in operations performed by Cloud Datastore. What should you do?

- A. Create the Key object for each Entity and run a batch get operation **Most Voted**
- B. Create the Key object for each Entity and run multiple get operations, one operation for each entity
- C. Use the identifiers to create a query filter and run a batch query operation
- D. Use the identifiers to create a query filter and run multiple query operations, one operation for each entity

Hide Solution

Discussion 37

Correct Answer: A 

Community vote distribution

A (100%)

Question #65

Topic 1

You need to upload files from your on-premises environment to Cloud Storage. You want the files to be encrypted on Cloud Storage using customer-supplied encryption keys. What should you do?

- A. Supply the encryption key in a .boto configuration file. Use gsutil to upload the files.
- B. Supply the encryption key using gcloud config. Use gsutil to upload the files to that bucket.
- C. Use gsutil to upload the files, and use the flag --encryption-key to supply the encryption key.
- D. Use gsutil to create a bucket, and use the flag --encryption-key to supply the encryption key. Use gsutil to upload the files to that bucket.

Hide Solution

Discussion 59

Correct Answer: A 

Community vote distribution

A (100%)

Your customer wants to capture multiple GBs of aggregate real-time key performance indicators (KPIs) from their game servers running on Google Cloud Platform and monitor the KPIs with low latency. How should they capture the KPIs?

- A. Store time-series data from the game servers in Google Bigtable, and view it using Google Data Studio.
- B. Output custom metrics to Stackdriver from the game servers, and create a Dashboard in Stackdriver Monitoring Console to view them. **Most Voted**
- C. Schedule BigQuery load jobs to ingest analytics files uploaded to Cloud Storage every ten minutes, and visualize the results in Google Data Studio.
- D. Insert the KPIs into Cloud Datastore entities, and run ad hoc analysis and visualizations of them in Cloud Datalab.

[Hide Solution](#)[Discussion](#) **83**

Correct Answer: B 

Reference:

<https://cloud.google.com/solutions/data-lifecycle-cloud-platform>

Community vote distribution

B (96%)



You have a Python web application with many dependencies that requires 0.1 CPU cores and 128 MB of memory to operate in production. You want to monitor and maximize machine utilization. You also want to reliably deploy new versions of the application. Which set of steps should you take?

A. Perform the following: 1. Create a managed instance group with f1-micro type machines. 2. Use a startup script to clone the repository, check out the production branch, install the dependencies, and start the Python app. 3. Restart the instances to automatically deploy new production releases.

B. Perform the following: 1. Create a managed instance group with n1-standard-1 type machines. 2. Build a Compute Engine image from the production branch that contains all of the dependencies and automatically starts the Python app. 3. Rebuild the Compute Engine image, and update the instance template to deploy new production releases.

C. Perform the following: 1. Create a Google Kubernetes Engine (GKE) cluster with n1-standard-1 type machines. 2. Build a Docker image from the production branch with all of the dependencies, and tag it with the version number. 3. Create a Kubernetes Deployment with the imagePullPolicy set to 'IfNotPresent' in the staging namespace, and then promote it to the production namespace after testing. **Most Voted**

D. Perform the following: 1. Create a GKE cluster with n1-standard-4 type machines. 2. Build a Docker image from the master branch with all of the dependencies, and tag it with 'latest'. 3. Create a Kubernetes Deployment in the default namespace with the imagePullPolicy set to 'Always'. Restart the pods to automatically deploy new production releases.

[Hide Solution](#)[Discussion](#) 103

Correct Answer: B 

Community vote distribution

C (72%)

A (28%)

Your company wants to start using Google Cloud resources but wants to retain their on-premises Active Directory domain controller for identity management. What should you do?

- A. Use the Admin Directory API to authenticate against the Active Directory domain controller.
- B. Use Google Cloud Directory Sync to synchronize Active Directory usernames with cloud identities and configure SAML SSO. **Most Voted**
- C. Use Cloud Identity-Aware Proxy configured to use the on-premises Active Directory domain controller as an identity provider.
- D. Use Compute Engine to create an Active Directory (AD) domain controller that is a replica of the on-premises AD domain controller using Google Cloud Directory Sync.

[Hide Solution](#)[Discussion](#) **74**

Correct Answer: B 

Community vote distribution

B (100%)

Question #69

Topic 1

You are running a cluster on Kubernetes Engine (GKE) to serve a web application. Users are reporting that a specific part of the application is not responding anymore. You notice that all pods of your deployment keep restarting after 2 seconds. The application writes logs to standard output. You want to inspect the logs to find the cause of the issue. Which approach can you take?

- A. Review the Stackdriver logs for each Compute Engine instance that is serving as a node in the cluster.
- B. Review the Stackdriver logs for the specific GKE container that is serving the unresponsive part of the application. **Most Voted**
- C. Connect to the cluster using gcloud credentials and connect to a container in one of the pods to read the logs.
- D. Review the Serial Port logs for each Compute Engine instance that is serving as a node in the cluster.

[Hide Solution](#)[Discussion](#) 70**Correct Answer:** B *Community vote distribution*B (100%)


Question #70

Topic 1

You are using a single Cloud SQL instance to serve your application from a specific zone. You want to introduce high availability. What should you do?

- A. Create a read replica instance in a different region
- B. Create a failover replica instance in a different region
- C. Create a read replica instance in the same region, but in a different zone **Most Voted**
- D. Create a failover replica instance in the same region, but in a different zone **Most Voted**

Hide Solution

Discussion **68**

Correct Answer: D 

Community vote distribution



Your company is running a stateless application on a Compute Engine instance. The application is used heavily during regular business hours and lightly outside of business hours. Users are reporting that the application is slow during peak hours. You need to optimize the application's performance. What should you do?

- A. Create a snapshot of the existing disk. Create an instance template from the snapshot. Create an autoscaled managed instance group from the instance template.
- B. Create a snapshot of the existing disk. Create a custom image from the snapshot. Create an autoscaled managed instance group from the custom image.
- C. Create a custom image from the existing disk. Create an instance template from the custom image. Create an autoscaled managed instance group from the instance template. **Most Voted**
- D. Create an instance template from the existing disk. Create a custom image from the instance template. Create an autoscaled managed instance group from the custom image.

[Hide Solution](#)[Discussion](#) 42

Correct Answer: C 

Community vote distribution

C (100%)

Question #72

Topic 1

Your web application has several VM instances running within a VPC. You want to restrict communications between instances to only the paths and ports you authorize, but you don't want to rely on static IP addresses or subnets because the app can autoscale. How should you restrict communications?

- A. Use separate VPCs to restrict traffic
- B. Use firewall rules based on network tags attached to the compute instances **Most Voted**
- C. Use Cloud DNS and only allow connections from authorized hostnames
- D. Use service accounts and configure the web application to authorize particular service accounts to have access

Hide Solution

Discussion 32

Correct Answer: B 

Community vote distribution

B (100%)

You are using Cloud SQL as the database backend for a large CRM deployment. You want to scale as usage increases and ensure that you don't run out of storage, maintain 75% CPU usage cores, and keep replication lag below 60 seconds. What are the correct steps to meet your requirements?

- A. 1. Enable automatic storage increase for the instance. 2. Create a Stackdriver alert when CPU usage exceeds 75%, and change the instance type to reduce CPU usage. 3. Create a Stackdriver alert for replication lag, and shard the database to reduce replication time.
- B. 1. Enable automatic storage increase for the instance. 2. Change the instance type to a 32-core machine type to keep CPU usage below 75%. 3. Create a Stackdriver alert for replication lag, and deploy memcache to reduce load on the master.
- C. 1. Create a Stackdriver alert when storage exceeds 75%, and increase the available storage on the instance to create more space. 2. Deploy memcached to reduce CPU load. 3. Change the instance type to a 32-core machine type to reduce replication lag.
- D. 1. Create a Stackdriver alert when storage exceeds 75%, and increase the available storage on the instance to create more space. 2. Deploy memcached to reduce CPU load. 3. Create a Stackdriver alert for replication lag, and change the instance type to a 32-core machine type to reduce replication lag.

[Hide Solution](#)[Discussion](#) 45

Correct Answer: A 

Community vote distribution

A (100%)

Question #74

Topic 1

You are tasked with building an online analytical processing (OLAP) marketing analytics and reporting tool. This requires a relational database that can operate on hundreds of terabytes of data. What is the Google-recommended tool for such applications?

- A. Cloud Spanner, because it is globally distributed
- B. Cloud SQL, because it is a fully managed relational database
- C. Cloud Firestore, because it offers real-time synchronization across devices
- D. BigQuery, because it is designed for large-scale processing of tabular data **Most Voted**

Hide Solution

Discussion 58

Correct Answer: D 

Reference:

<https://cloud.google.com/files/BigQueryTechnicalWP.pdf>

Community vote distribution

D (92%) 8%

Question #75

Topic 1

You have deployed an application to Google Kubernetes Engine (GKE), and are using the Cloud SQL proxy container to make the Cloud SQL database available to the services running on Kubernetes. You are notified that the application is reporting database connection issues. Your company policies require a post-mortem. What should you do?

- A. Use `gcloud sql instances restart`.
- B. Validate that the Service Account used by the Cloud SQL proxy container still has the Cloud Build Editor role.
- C. In the GCP Console, navigate to Stackdriver Logging. Consult logs for (GKE) and Cloud SQL. **Most Voted**
- D. In the GCP Console, navigate to Cloud SQL. Restore the latest backup. Use `kubectl` to restart all pods.

Hide Solution

Discussion 60

Correct Answer: C 

Community vote distribution

C (100%)

Question #76

Topic 1

Your company pushes batches of sensitive transaction data from its application server VMs to Cloud Pub/Sub for processing and storage. What is the Google-recommended way for your application to authenticate to the required Google Cloud services?

- A. Ensure that VM service accounts are granted the appropriate Cloud Pub/Sub IAM roles. **Most Voted**
- B. Ensure that VM service accounts do not have access to Cloud Pub/Sub, and use VM access scopes to grant the appropriate Cloud Pub/Sub IAM roles.
- C. Generate an OAuth2 access token for accessing Cloud Pub/Sub, encrypt it, and store it in Cloud Storage for access from each VM.
- D. Create a gateway to Cloud Pub/Sub using a Cloud Function, and grant the Cloud Function service account the appropriate Cloud Pub/Sub IAM roles.

Hide Solution

Discussion 36

Correct Answer: A 

Community vote distribution

A (100%)

Question #77

Topic 1

You want to establish a Compute Engine application in a single VPC across two regions. The application must communicate over VPN to an on-premises network. How should you deploy the VPN?

- A. Use VPC Network Peering between the VPC and the on-premises network.
- B. Expose the VPC to the on-premises network using IAM and VPC Sharing.
- C. Create a global Cloud VPN Gateway with VPN tunnels from each region to the on-premises peer gateway.
- D. Deploy Cloud VPN Gateway in each region. Ensure that each region has at least one VPN tunnel to the on-premises peer gateway. **Most Voted**

Hide Solution

Discussion 44

Correct Answer: D 

Community vote distribution

D (100%)

Question #78

Topic 1

Your applications will be writing their logs to BigQuery for analysis. Each application should have its own table. Any logs older than 45 days should be removed. You want to optimize storage and follow Google-recommended practices. What should you do?

- A. Configure the expiration time for your tables at 45 days
- B. Make the tables time-partitioned, and configure the partition expiration at 45 days **Most Voted**
- C. Rely on BigQuery's default behavior to prune application logs older than 45 days
- D. Create a script that uses the BigQuery command line tool (bq) to remove records older than 45 days

Hide Solution

Discussion 54

Correct Answer: B 

Community vote distribution

B (100%)

You want your Google Kubernetes Engine cluster to automatically add or remove nodes based on CPU load. What should you do?

- A. Configure a HorizontalPodAutoscaler with a target CPU usage. Enable the Cluster Autoscaler from the GCP Console. **Most Voted**
- B. Configure a HorizontalPodAutoscaler with a target CPU usage. Enable autoscaling on the managed instance group for the cluster using the gcloud command.
- C. Create a deployment and set the maxUnavailable and maxSurge properties. Enable the Cluster Autoscaler using the gcloud command.
- D. Create a deployment and set the maxUnavailable and maxSurge properties. Enable autoscaling on the cluster managed instance group from the GCP Console.

[Hide Solution](#)[Discussion](#) **62**

Correct Answer: A 

Community vote distribution

A (100%)

You need to develop procedures to verify resilience of disaster recovery for remote recovery using GCP. Your production environment is hosted on-premises. You need to establish a secure, redundant connection between your on-premises network and the GCP network.

What should you do?

- A. Verify that Dedicated Interconnect can replicate files to GCP. Verify that direct peering can establish a secure connection between your networks if Dedicated Interconnect fails.
- B. Verify that Dedicated Interconnect can replicate files to GCP. Verify that Cloud VPN can establish a secure connection between your networks if Dedicated Interconnect fails. **Most Voted**
- C. Verify that the Transfer Appliance can replicate files to GCP. Verify that direct peering can establish a secure connection between your networks if the Transfer Appliance fails.
- D. Verify that the Transfer Appliance can replicate files to GCP. Verify that Cloud VPN can establish a secure connection between your networks if the Transfer Appliance fails.

[Hide Solution](#)[Discussion](#) 59

Correct Answer: B 

Community vote distribution

B (100%)